

MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
PROCESS VESSEL COLUMN INTEGRITY & GAMMA SCAN REPORT - CDU-1

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Equipment ID: MRPL-COL-301

Equipment Description: Main Atmospheric Crude Distillation Column

Operating Unit: Crude Distillation Unit 1 (Capacity 12.0 MMTPA)

Inspection Date: 2026-09-06

Radiometric Inspection Contractor: Tracerco Industrial Radiometrics Ltd.

HYDRAULIC & RADIOMETRIC SCAN SUMMARY (TRAYS 15 TO 32):

- Total Column Differential Pressure (Top to Bottom): 0.78 bar (SOP Maximum: 0.65 bar).
- Tray 20 to 24 Differential Pressure: 0.38 bar (High liquid hold-up / flooding).
- Gamma Radiometric Density Scan: Detected severe aerated liquid backup on Trays 21-22.
- Kerosene/Diesel Fractionation Cut: ASTM D86 95% boiling point overlap widened by 22 deg C.
- **Column Top Temperature: 134 deg C (Design Operating Window: 110 - 122 deg C).**
- Reflux Accumulator Level: Surging between 35% and 88% with heavy entrainment.

DIAGNOSIS & ROOT CAUSE:

Localized downcomer choking and tray damage around wash zone (Trays 21-23) triggered by sudden water flashing from crude feed tank bottom settling failure.

MAINTENANCE RECOMMENDATION & MITIGATION:

1. Immediate adjustment of pumparound duty (+15%) to de-inventory middle wash section.
2. Reduce crude furnace outlet temperature by 4 deg C to reduce vapor superficial velocity.
3. Schedule mini-turnaround internal entry to replace collapsed valve trays (Trays 21-23).
4. Inspect chimney tray seals and downcomer bolting with austenitic SS 316L hardware.